

Adaalat AI — Pakistan’s Multilingual AI Legal Assistant

Project Title

Adaalat AI: An AI-Powered Multilingual Legal Assistance Platform for Pakistani Citizens

1. Introduction

Access to legal information remains a major challenge in Pakistan due to high legal consultation costs, lack of awareness, language barriers, and the complexity of legal procedures. Most citizens do not fully understand their legal rights in matters such as tenant disputes, family conflicts, labor exploitation, inheritance issues, or police complaints. Legal resources are usually written in complex English legal terminology, making them inaccessible to a large portion of the population.

In many cases, people suffer injustice not because laws do not exist, but because they do not understand how those laws protect them. Existing legal platforms either provide static information or only connect users with lawyers without offering simplified legal guidance.

Adaalat AI is proposed as a multilingual AI-powered legal assistance platform designed specifically for Pakistan. The system provides conversational legal guidance in Urdu, Roman Urdu, and English using Artificial Intelligence and Retrieval-Augmented Generation (RAG). It helps users understand Pakistani laws in simple language through text and voice interaction.

The system is designed to bridge the gap between ordinary citizens and legal awareness by providing accessible, affordable, and reliable legal assistance.

2. Problem Statement

Pakistan faces a major legal accessibility problem where many citizens cannot understand or afford legal assistance. Legal information is often written in complex English and difficult legal terminology, making it inaccessible to a large portion of the population. Many people facing issues such as tenant disputes, unpaid salaries, family conflicts, or police complaints remain unaware of their legal rights and procedures. Existing platforms either provide static legal information or lack conversational support in local languages. Therefore, there is a strong need for a multilingual AI-powered legal assistance system that provides simplified legal guidance in Urdu, Roman Urdu, and English using authentic Pakistani legal sources.

3. Executive Summary

Adaalat AI is a multilingual AI-powered legal assistance platform developed specifically for Pakistan to improve public access to legal knowledge and awareness. The platform acts as a virtual legal advisor that helps users understand their legal rights through conversational interaction in Urdu, Roman Urdu, and English.

The system uses Retrieval-Augmented Generation (RAG) to ensure responses are generated from authentic Pakistani legal sources rather than generic AI-generated information. This improves the reliability and accuracy of legal guidance.

Adaalat AI simplifies legal concepts into understandable language and provides guidance for common legal matters such as FIR filing, labor rights, family disputes, tenant conflicts, and inheritance issues. The platform also includes legal document generation, lawyer recommendations, anonymous consultation, and court reminder functionalities.

The project aims to make legal awareness accessible to every citizen regardless of education level, language, or financial status.

4. Motivation

Millions of Pakistanis face legal problems without understanding their rights because legal knowledge is difficult to access and expensive professional consultation is often unaffordable. Rural communities, low-income groups, and less educated citizens are especially affected by this issue.

Adaalat AI is motivated by the belief that justice and legal awareness should not be limited by language, literacy, or financial barriers. Modern Artificial Intelligence technologies now make it possible to simplify legal systems and deliver understandable legal guidance directly to citizens in their preferred language.

The project aims to empower people with legal awareness and help them make informed decisions regarding their rights and legal procedures.

5. Objectives

Primary Objectives

- Develop a multilingual AI legal assistant for Pakistani citizens
- Provide legal guidance in Urdu, Roman Urdu, and English

- Simplify complex Pakistani laws into understandable language
- Increase legal awareness among the public
- Provide instant legal assistance through conversational AI

Secondary Objectives

- Generate basic legal documents automatically
- Provide step-by-step FIR filing guidance
- Assist users in family, labor, and tenant law matters
- Connect users with verified lawyers when professional consultation is required
- Provide legal reminders and notifications

6. Existing System / Competitor Analysis

Qanoon.pk

Provides legal articles and legal information but lacks AI-based conversational assistance and multilingual interaction.

Moakil

Focuses mainly on connecting users with lawyers and does not provide intelligent legal guidance or simplified explanations.

LegalWiz

Offers legal documentation services but is not specifically designed for Pakistani legal systems or multilingual accessibility.

DoNotPay

An advanced AI legal assistant developed for Western legal systems and not applicable to Pakistani laws and procedures.

7. Research Gap

Current legal platforms in Pakistan lack:

- Multilingual AI-based legal interaction
- Simplified explanations of Pakistani laws
- Retrieval-based law-grounded responses
- Legal assistance in Urdu and Roman Urdu
- AI-powered legal document generation
- Citizen-focused legal awareness systems

Adaalat AI addresses these limitations by combining AI, multilingual NLP, and Pakistani legal knowledge retrieval into a single accessible platform.

8. Proposed Solution

Adaalat AI will function as an intelligent digital legal assistant that allows users to ask legal questions through text or voice input. The system processes the query and retrieves relevant legal information from Pakistani legal documents using a RAG-based architecture.

The retrieved information is then used by the AI model to generate simplified legal guidance in Urdu, Roman Urdu, or English. The platform also provides document generation, lawyer recommendations, court reminders, and legal awareness modules.

The system is designed to improve legal understanding while ensuring responses remain grounded in authentic legal data.

9. Functional Requirements

- User registration and login system
- Multilingual chatbot support
- Voice and text-based legal interaction
- RAG-based legal knowledge retrieval
- Legal document generation
- FIR filing guidance system
- Family law support
- Labor law guidance
- Tenant rights assistance
- Lawyer recommendation module
- Anonymous consultation mode
- Court date reminder system
- Legal updates and awareness notifications

10. Non-Functional Requirements

- High system availability
- Fast response time
- Secure authentication and user privacy
- Accurate law-grounded responses
- Scalable architecture for multiple users
- User-friendly and accessible interface
- Cross-platform compatibility

11. Key Features

Multilingual Legal Chatbot

Provides legal guidance in Urdu, Roman Urdu, and English.

Legal Document Generator

Generates affidavits, notices, and agreements automatically.

FIR Guidance System

Provides step-by-step instructions for FIR filing procedures.

Family Law Support

Offers guidance regarding divorce, khula, inheritance, and child custody.

Labor Rights Assistance

Helps users understand workplace rights and salary disputes.

Tenant Rights Module

Provides guidance regarding rental agreements and eviction disputes.

Lawyer Finder

Connects users with verified legal professionals.

Anonymous Consultation Mode

Allows users to ask sensitive legal questions privately.

Court Reminder System

Sends reminders regarding hearings and appointments.

12. System Architecture

Frontend

- Flutter Mobile Application

Backend

- FastAPI Framework

AI Layer

- LangChain-based RAG Pipeline

Vector Database

- ChromaDB

Main Database

- PostgreSQL

Speech Processing

- Whisper Speech-to-Text

Large Language Model

- OpenAI API / Open-Source LLM

Authentication

- JWT Authentication System

13. System Workflow

User Query → Voice/Text Input → Speech-to-Text Processing → Query Processing → Legal Data Retrieval from ChromaDB → LLM Response Generation → Simplified Legal Guidance → User Output

14. AI / Machine Learning Components

- Retrieval-Augmented Generation (RAG)
- Urdu Natural Language Processing (NLP)
- Embedding-based semantic search
- Legal category classification
- Speech-to-text processing
- Context-aware response generation

15. Technology Stack

Component	Technology
Frontend	Flutter
Backend	FastAPI
Database	PostgreSQL
Vector Database	ChromaDB
AI Framework	LangChain
LLM	OpenAI / Llama
Voice Processing	Whisper
Authentication	JWT

16. Expected Outcomes

- Improved public legal awareness
- Easier understanding of Pakistani laws
- Increased accessibility to legal guidance
- Reduced dependency on expensive consultations for basic issues
- Better support for low-literacy and rural populations
- Promotion of digital legal accessibility in Pakistan

17. Future Enhancements

- Integration with government legal portals
- Real-time lawyer consultation system
- AI-powered case prediction
- Offline support for rural users
- Additional regional language support
- Court case tracking system

18. Conclusion

Adaalat AI is a socially impactful and technically advanced Final Year Project designed to improve legal accessibility and awareness in Pakistan. By combining Artificial Intelligence, multilingual language processing, and Retrieval-Augmented Generation (RAG), the system transforms complex Pakistani laws into understandable legal guidance for ordinary citizens.

The platform does not replace lawyers; instead, it empowers citizens with legal knowledge and helps them understand their rights, legal procedures, and when professional legal assistance is required. Adaalat AI aims to bridge the gap between justice and accessibility by making legal awareness available to everyone regardless of language, literacy, or financial status.